

Julian Andres Gomez Niño

SYSTEMS AND COMPUTER ENGINEERING STUDENT

Bogotá, Colombia

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Summary

Systems and Computer Engineering student at the National University of Colombia with experience building scalable and data-driven software systems. Strong background in Python, SQL, backend development, and database design, with particular interest in Artificial Intelligence and applied machine learning systems. Experienced working in collaborative environments developing reliable solutions focused on performance, scalability, and maintainability. Proactive problem-solver with strong analytical thinking and ownership mindset.

Education

National University of Colombia

Bogotá, Colombia

B.S. IN SYSTEMS AND COMPUTER ENGINEERING

Mar. 2022 - Present

- Developing strong foundations in Artificial Intelligence, Machine Learning, and Data Science.
- Engaged in academic and volunteer initiatives related to technology and management.

School RR Oblatas al Divino Amor

Bogotá, Colombia

ACADEMIC BACCALAUREATE

Feb. 2015 - Nov. 2020

- Graduated with focus on mathematics and computer fundamentals.

Projects

UNxchange

Github: UNxchange

DISTRIBUTED UNIVERSITY EXCHANGE PLATFORM BASED ON MICROSERVICES ARCHITECTURE

2025

- Collaborated in the development of a distributed university exchange platform using FastAPI, Go, React, and TypeScript.
- Contributed to backend services implementing JWT/OAuth2 authentication, GraphQL/REST APIs, and PostgreSQL/MongoDB integrations.
- Developed frontend features including protected routes, modular components, and role-based workflows.
- Participated in the integration of centralized API gateway routing, Prometheus monitoring, Dockerized deployments, and automated E2E testing with pytest and Selenium.
- Supported the development of analytics services and KPI endpoints for institutional exchange agreement reporting.

ElGamal Secure Voting System

Github: ElGamal Voting System

SECURE WEB VOTING PLATFORM WITH ELGAMAL ENCRYPTION AND ANONYMIZED AUDIT TRAIL

2025

- Collaborated in the development of a secure voting platform using Flask and SQLAlchemy, focusing on voter anonymity and election integrity.
- Implemented features related to encrypted ballot storage, vote hashing, and one-vote validation constraints at the database level.
- Contributed to the integration of custom ElGamal cryptography workflows with per-election key generation and administrative decryption processes.
- Developed and validated automated tests covering encryption/decryption correctness, integrity checks, and election scheduling logic.

CarWash Management Platform

Github: CarWash

FULL-STACK SERVICE-MANAGEMENT PLATFORM WITH SECURE REPORTING WORKFLOWS

2024

- Collaborated in the development of a full-stack service-management platform using Node.js, Express, MongoDB, React, and Netlify.
- Contributed to REST API development, including validation logic, state management workflows, and historical audit tracking.
- Implemented JWT-based authentication middleware, credential protection with bcrypt, and secure access-control mechanisms.
- Developed reporting and analytics features using server-side aggregation and PDF generation for operational metrics.
- Integrated frontend data workflows, routing, and production-style deployment configurations.

Skills

TECHNICAL SKILLS

Programming Python, Java, C++, JavaScript, R, SQL

Libraries NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Tensorflow, Pytorch, FastAPI, Typescript

Tools Excel, PowerBI, Appian

INTERPERSONAL SKILLS

Team collaboration, Communication, Problem-solving, Curiosity, Analytical thinking

LANGUAGES

Spanish Native

English Advanced (C1 EF SET)

German Basic

Certificates

Certified AI Practitioner AWS	Sep 2026 - Sep 2029
Data Analysis with Python IBM	Sep 2026
Claude Certified Architect Prep DEVCOMPASS	Jul 2026
Databases and SQL for Data Science with Python IBM	Feb 2026
Python Project for Data Science IBM	Sep 2025
Introduction to Statistical Analysis: Hypothesis Testing SAS	May 2025

A COMPLETE LIST OF CERTIFICATIONS IS AVAILABLE ON MY LINKEDIN PROFILE

Extracurricular Activities

IEEE Technology & Engineering Management Society (TEMS UN) SECRETARY & MEMBER	Bogotá, Colombia Mar. 2022 – Feb. 2026
- Coordinated administrative and communication activities within the student chapter. - Supported the organization of events focused on engineering and technology management. - Encouraged student participation in interdisciplinary and professional development initiatives.	
Generative Models Research Seminar MEMBER	Bogotá, Colombia Oct. 2025 – Present
- Continuous learning and exploration of different types of generative models. - Developed and redesigned the official website of the seminar.	
IEEE Computer Society MEMBER	Bogotá, Colombia Mar. 2025 – Feb. 2026
Participated in technical outreach activities and the promotion of best practices in computer science.	
UNAL Finance Club (FIN-UN) MEMBER	Bogotá, Colombia Oct. 2024 – Aug. 2026
Contributed to the analysis of financial and economic topics relevant to university management and student initiatives.	